MIRAI INNOVATION RESEARCH INSTITUTE · OSAKA

# VR EEG Studio

Measuring brain activity while people use virtual reality, so we can test VR experiences in real places, not only in the lab.

Field-test proposal

Prepared for TEQS

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WHY FIELD TESTS

# VR is used in shops, clinics and public spaces. Our measurements should come from there too.

### What we measure

Mental effort and relaxation, from the EEG of 5 forehead electrodes, second by second.

### While doing what

Short VR tasks: buying from a virtual vending machine, or a guided relaxation scene.

### Why outside the lab

Real users, real noise and real time limits show whether the experience works where it will be used.

HOW IT WORKS

# One laptop joins the brain signal and the game events

1 · EEG HEADSET

## WAVEX

Wireless EEG. We use its 5 frontal channels: Fp1, Fp2, F3, Fz, F4.

2 · VR GAME

## Headset + game

Sends an event each time something happens, e.g. “item selected”.

3 · VR EEG STUDIO (LAPTOP)

## Syncs, cleans and checks

- Same clock for EEG and game events (LSL)
- Filters noise automatically
- Shows electrode quality live
- Saves to disk every second

4 · RESULTS

## Files + report

Standard XDF data files, a one-page report per session and study statistics.

Runs offline on one laptop. No internet or cloud service is needed at the site.

ONE PARTICIPANT, STEP BY STEP

# About 30–40 minutes per person

~10 min
**Prepare**
Consent, headset on, electrodes green

1–2 min
**Rest**

10–15 min
**VR tasks**
2–3 trials of 5 min with short breaks

~5 min
**Wrap-up**
Headset off, report

The app guides the operator through each step; timers stop and save by themselves.

[2] people from our team run each session: one operator, one with the participant.

With one setup, we can see about 8–10 people in a day.

Times are planning estimates from our current protocols; they can be shortened for busy sites.

WHAT PARTICIPANTS DO

# Two short VR experiences

## VR Vending Machine

The participant buys the items shown on a screen from a virtual vending machine.

| Step | Time |
|---|---|
| Rest, eyes open | 1 min |
| Task trials | 3 × 5 min |
| Breaks | 30 s each |

**Question: how much mental effort does the task take, and does it drop with practice?**

## VR Relaxation

The participant looks around a calm scene and breathes slowly.

| Step | Time |
|---|---|
| Rest, eyes closed | 2 min |
| Relaxation scenes | 2 × 5 min |
| Breaks | 1 min |

**Question: does the scene help people relax compared with their own rest?**

Seated, no walking. Participants can stop at any moment.

FOR HOST SITES

# We bring everything. A site gives us a corner and people.

## What we bring

- EEG headset, VR headset and laptop
- Trained staff to run every session
- Consent forms in Japanese
- Cleaning of all equipment between people

## What we need

- A quiet space of about 3 × 3 m
- A table, two chairs and one power outlet
- Away from large electrical equipment
- Adult volunteers: [__] people over [__] days

Examples of good sites: [showrooms, community centres, clinics, universities, offices]

SAFETY AND PRIVACY

# Non-invasive, voluntary and anonymous

### Only reads signals

The headset sends no current to the body; it only listens.

### Written consent

Every participant signs first and can stop at any time, without giving a reason.

### Codes, not names

Data is stored under a participant code on our laptop, not in the cloud.

### Research, not diagnosis

We report group trends; no one gets a medical result.

VR sessions are short and seated to limit motion sickness; staff stay beside the participant.

NEXT STEPS WITH TEQS

# Let’s find the first site together

01

### Shortlist

TEQS suggests 2–3 candidate sites.

02

### Visit and demo

We show the system to the site and check the space.

03

### Pilot days

A first test, then a summary report shared with the site.

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